When Retrieval Succeeds and Reasoning Fails: Multi-Hop RAG on the FRAMES Benchmark

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Abstract

Retrieval-Augmented Generation (RAG) has emerged as a leading approach for grounding large language models (LLMs) in external knowledge. The effectiveness of these models on complex multi-hop questions, where answering correctly requires chaining evidence across several documents, still remains an open question. In this work, we evaluate six different models of increasing complexity and sophistication on the FRAMES benchmark dataset, a challenging 824-question dataset designed to test factuality, retrieval, and multi-hop reasoning. We use two extractive baselines (RoBERTa-SQuAD with and without BM25 retrieval, a 14-billion-parameter instruction-tuned LLM (Qwen2.5-14B-Instruct) in a zero-shot and few-shot setting, to serve as our baselines. Additionally, we test two RAG configurations, a single-step and iterative three-hop retrieval. These models are built on a Wikipedia corpus of 3,963 articles indexed with BM25. Using an LLM-as-judge evaluation protocol we find that single-step RAG nearly doubles the accuracy over the zero-shot LLM baseline, and that iterative multi-hop retrieval further improves both retrieval and answer accuracy, confirming that retrieval is the most impactful single intervention on this benchmark. Error analysis also suggests that reasoning failure, and not retrieval failure, is the most dominant bottleneck. Even when at least one gold article is successfully retrieved, our best model still answers incorrectly 69.3% of the time, with post-processing and numerical questions proving to be the hardest across all conditions. These findings highlight that improving retrieval alone is insufficient, and that reasoning over retrieved evidence remains the primary challenge for multi-hop RAG systems.

1 Introduction

Large Language Models have become impressively capable at answering questions and reasoning, yet their reliability is still bounded by fundamental limitations. Firstly, they only know the information on which they have been trained. Any event that has occurred after training is not accessible to the model. Secondly, LLMs have been shown to struggle to access information that is rarely mentioned in their training corpus (Kandpal et al., 2023). Both limitations can lead to false answers built on hallucinations.

Retrieval Augmented Generation is a technique that mitigates these problems by providing the models access to external documents that are retrieved based on a specific question asked by the user. In this project, we will be using the FRAMES benchmark to evaluate how RAG-based models compare against a baseline model and an LLM in a zero-shot and a few-shot setting. The outputs will be evaluated on traditional evaluation metrics and by an instructed LLM acting as a judge. This will give a clear picture of where the model fails and how it can be improved in the future.
This paper addresses the following research questions:

RQ1. To what extent does a RAG-based system improve question answering performance?

RQ2. How do generative LLMs compare to fine-tuned models?

RQ3. What factors negatively affect retrieval performance?

Our results show that single-step RAG nearly doubles the accuracy over a zero-shot LLM baseline, and that three-hop iterative retrieval further improves retrieval substantially, also answer accuracy.

This paper is structured as follows. Section 2 reviews related work on RAG-based models and similar datasets. Section 3 describes the dataset we will be using to test our models on. Section 4 explains how the models and methods were implemented. Section 5 explains what metrics we

have chosen to evaluate our models on and section 6 shows the results of these evaluations. Section 7 provides an error analysis, explaining the type of errors the models make and how they can be mitigated in the future. Section 8 discusses the limitations of our work and how it could be better. Section 9 tackles the ethical consideration and section 10 concludes the paper.

2 Related Work

2.1 Retrieval Augmented Generator

The concept of augmenting language models with external retrieval was first formalized by Patrick Lewis and other researchers from Meta AI Research (Lewis et al., 2021), in a paper called "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks". Lewis and his team introduced the first RAG-model by combining a pre-trained neural retriever with a seq-2-seq generator. Their paper concluded that RAG-models generate more specific, diverse and accurate answers compared to a state-of-the-art parametric-only seq2seq baseline. Our work builds directly on the foundation, exploring whether retrieval improves generated outputs by testing it on the challenging FRAMES benchmark.

2.2 HotpotQA

The FRAMES dataset we will be using in our project is largely inspired by HotpotQA, one of the first multi-hop benchmarks that requires reasoning across multiple documents (Yang et al., 2018). This dataset was not developed to specifically stress test RAG-systems and mainly included bridge questions across a maximum of two documents.

2.3 LLM as a judge

Evaluating generated answers given by LLMs with traditional evaluation metrics can be a difficult task. (Es et al., 2025) presents a way to evaluate answers by incorporating an LLM-as-judge approach into the evaluation pipeline. We decided to use this in our project.

2.4 BasicMultiHop

(Khattab et al., 2023). BasicMultiHop implements the same iterative retrieval pattern as one of our models in this project. Our C2 model is essentially a manual implementation of BasicMultiHop. An alternative to our approach can be found in (Asai et al., 2023). This uses adaptive retrieval reflection tokens rather than a fixed 3-hop.

2.5 Lost in the Middle

(Liu et al., 2023). This paper explains the U-shaped accuracy-by-position curve and why the ordering of passages matters in multi-document contexts.

2.6 State-of-the-art models on the topic

In the original FRAMES white-paper, (Krishna et al., 2025) the team tested several state-of-the-art models on the dataset. They used Gemini-Pro-1.5 as the primary model under evaluation and as a LLM judge to evaluate the correctness of the generated answers. Their baseline was naive prompting on a non-retrieval model that achieved 40.8% accuracy. The accuracy rose to 66% after implementing a multi-step retrieval pipeline and the best results were 72.9% accuracy under oracle retrieval conditions with a reasoning failure rate of around 27%. Our setup differs from Krishna et al. (2025) in model choice and retrieval pipeline, making direct comparison approximate, but still useful.

3 Dataset

The dataset we will be using to evaluate our model is called FRAMES (Factuality, Retrieval, And reasoning Measurement Set) and is developed by Google. It is an evaluation set consisting of 824 questions designed to test the factuality, retrieval and reasoning of RAG systems. Each row of the dataset includes a challenging multi-hop question that requires information from 2-15 Wikipedia articles to be answered. The questions spans multiple topics including; history, sports, music and science. Each question is labeled with a reasoning type that explains what type of question it is. These reasoning types include: numerical, tabular, temporal, multiple constraints and post-processing. The Wikipedia articles are ordered in the sequence that you would need to reason through the question step-by-step. The average length of each question is 27.6 words (ranging from 8 to 130 words). In contrast, the answers are much smaller with an average length of 5.2 words (ranging from 1 to 260 words). Each question has an average of 3.2 links attached.

A possible limitation for this dataset could be the source of the retrieved data. Wikipedia articles can be written by anyone as long as they are accepted by their moderators. However, human-annotators

from Google completed quality checks on the data to ensure that the provided answer was correct and could be answered using the associated Wikipedia articles. The only concern is that Wikipedia updates their articles overtime, which could result in missing information.

4 Methods/Experiments

4.1 Wikipedia Corpus Construction

Gold articles: To construct a corpus of Wikipedia articles that the retriever would pick from for each query we first extracted all the unique Wikipedia articles from the FRAMES wiki_links columns across all the 824 questions. We identified 2,507 unique gold articles, which were then fetched with each article's full-text content. Some articles failed on the first attempt and were attempted to be fetched once more. Finally, we ended up with 2,465 articles successfully fetched, meaning only a few titles were permanently left out of the corpus.

Distractor articles: Including gold articles, we also added 1,500 random Wikipedia articles to act as distractors. These are fetched to simulate a more realistic retrieval corpus, where not every article is relevant for any question. The rationale behind this is that a retrieval from a corpus that only contains gold articles would be unrealistically easy and not representative of the challenges a retrieval system would face in a real world setting. ((Cuconasu et al., 2024) Power of Noise gives evidence that corpus composition significantly affects RAG accuracy).

The final corpus contains 3,963 articles in total, stored as corpus.json.

4.2 BM25 Index

After creating a Wikipedia corpus we ran a BM25 index over all the 3,963 article texts. This is done on lowercase whitespace split on the full article content, using no stemming or stop-word removal. For the passage extraction of the extractive models (B2, C1, C2), we used a sliding window over the articles content of 500 character chunks and a 250-character stride. For B2 we used the top-1 passage per article, with 5 articles concatenated, and then truncated to 3,000 characters in total. For C1 and C2 we used the top-3 passages per articles, and 700 characters per passage for the final answer context.

(Karpukhin et al., 2020) presents a way to use embeddings so that retrieval is based on meaning rather than matching exact words. However, this approach would require in-domain training and we therefore deem it unnecessary for the experiments we will conduct.

4.3 Retrieval Evaluation

We evaluated BM25 Recall@k on all the 824 questions. Recall@k definition for a question with gold articles set G and retrieved set R@k:

$$\text{Recall@}k = \frac{|G \cap R_k|}{|G|}$$

Results averaged over all questions:

k	Recall@k
1	0.212
3	0.393
5	0.449
10	0.508

Table 1: BM25 Recall@k evaluated on all 824 FRAMES questions.

Required articles	Recall@5	n questions
2	0.527	310
3	0.453	288
4	0.354	134
5	0.378	37
6	0.254	19
7	0.297	13
8	0.438	6
9	0.259	3
10	0.167	3
11	0.198	11

Table 2: BM25 Recall@5 broken down by the number of gold Wikipedia articles required to answer each question

The results suggest that there clearly is a degradation trend with more required articles (also take into account that small n questions can lead to noise). This explains why single-step retrieval might be insufficient for multi-hop FRAMES questions, and more complex methods such as a hybrid retriever might be desired. As a reference point the FRAMES paper reported per-hop BM25 recall of 0.12-0.15 in their multi-step setting. Our Recall@5 at 0.449 is from a single query over the full question.

4.4 Baseline Models

All baselines are run on the same 412-question stratified subset.

B1: BERT Extractive - No retrieval Model: deepset/roberta-base-squad2, which is a RoBERTa-base that is fine-tuned on SQuAD2. Context passed to the model: The question text. The rationale here is to find the absolute floor, and prove that an extractive span model with no retrieval cannot answer complex multi-hop questions requiring external knowledge. Expected to score very near 0.

B2: BM25 + BERT Extractive Model: deepset/roberta-base-squad2 (Same as B1) Retrieval: BM25 top-5 articles. Context passed to the model: 5 passages concatenated, and truncated to 3,000 characters. This is included to compare how much the RoBERTa model improves when given context using BM25 retrieval. It is expected to be weak on FRAMES because extractive models can't synthesize across multiple documents, and that correct answers often require inference, which this model can't do.

B3: LLM Zero-Shot Model: Qwen/Qwen2.5-14B-Instruct No retrieval, so it relies entirely on the model's pre-trained knowledge. This establishes what a capable LLM knows about the FRAMES questions already without any grounding.

B4: LLM-Few-Shot Model: Qwen/Qwen2.5-14B-Instruct Given 3 few-shot examples. The few-shot example questions are not as complex as the FRAMES questions, but the example answers are concise and generally in the same style as the gold answers are in FRAMES, however even the FRAMES dataset we can find inconsistencies where some answers are extremely short and concise, while others are much longer and in free-form. This baselines tests if format conditioning helps the LLM produce more appropriate short-form answers.

4.5 Core Neural Models

Both RAG models use the same Qwen2.5-14B-Instruct as the generator, which is also the same model used in the B3 and B4 baselines.

C1: Single-Step RAG This model is divided into three steps. First it does BM25 retrieval with k=5 and the query is the original question. Then for the context it gets the top-3 passages per articles, on 700 chars each. Finally the LLM generates an answer from the context. This is the "Naive RAG" paradigm (Gao et al., 2024), which is linear retrieve-then-generate.

C2: Multi-Step RAG (3 hops) This model has an iterative retrieval loop on 3 hops total. In hop 1 the query is the original question, and BM25 with k=5 accumulates articles. In hop 2 the LLM generates a new search query given the question, accumulated context thus far, and a list of previous queries. BM25 with k=5 then adds new unique articles. Finally hop 3 is the same as hop 2, where it finally will be instructed to answer the original question. We implemented deduplication, which means articles already in the retrieved set are skipped. This allows the model to see up to 15 articles. This is the "Iterative RAG" / "Modular RAG" paradigm (Gao et al., 2024), which is the same pattern as FRAMES's own multi-step baseline (Krishna et al., 2025) and DSPy's BasicMultiHop module (Khattab et al., 2023).

We can see from the C1 to the C2 model the retrieval recall is substantially improved, and would likely also be further improved if a model with more hops was tested. Multi-hop retrieval increases recall by 17.9%, which confirms that iterative query generation helps surface articles that the original question alone would miss.

Model	Retrieval Recall
C1: Single-hop RAG	0.437
C2: Multi-hop RAG	0.616

Table 3: Retrieval recall for the two RAG models

5 Evaluation

Token F1 and Exact Match: In order to evaluate this properly we used normalization where we put all inputs in lowercase, and stripped punctuation. As for Token F1 we used bag-of-tokens overlap, harmonic mean of per-token precision and recall. The Exact Match (EM) metric will be 1 if normalized strings are identical, if not it will return 0. These are traditional metrics for evaluation, but not ideal for getting a reliable understanding of model performance. Responses from LLMs are verbose, and many questions are in free-form. We expect EM to be near 0 for all LLM models. We also don't expect Token F1 to score well on any of the models.

LLM-as-Judge: The primary evaluation metric for all models will be LLM-as-Judge. The models used as judge is the Qwen2.5-14B-Instruct (same model used for generation). The judge model is given the question, gold answer, and predicted answer, and will output either CORRECT or INCORRECT. A response is CORRECT if the correct fac-

tual answer appears in the text, even within an explanation (which often might be the case with LLMs). A response is INCORRECT if it never states the correct answer, states the wrong fact, refuses to answer, or hedges its response without committing to a single answer. Something to note is that using the same model as generator and judge is a limitation. We will speak more about this in Limitations.

This evaluation design mirrors RAGAS (Es et al., 2025), which uses LLM-based metrics to diagnose RAG pipeline components without the reference answers, and also the FRAMES paper itself which uses Gemini Pro 1.5 as a judge.

6 Results

6.1 Main Results Table

Evaluated on 412 questions (50% stratified subset, seed 42):

Model	LLM-Judge	F1	EM	R.Recall
B1	0.005	0.063	0.000	—
B2	0.068	0.065	**0.032**	0.437
B3	0.138	**0.145**	0.044	—
B4	0.187	0.080	0.002	—
C1	0.265	0.107	0.002	0.437
C2	**0.294**	0.120	0.002	**0.616**

Table 4: Main results on the 412-question stratified subset.

Key observations:

B1 with an accuracy of 0.5% confirms that an extractive model that has no retrieval can't answer FRAMES questions. From manual review, the 0.5% are from lucky extractions from the question text itself.

B2 shows that BM25 retrieval for extractive models helps, but it is still fundamentally limited because the extractive model is not suited for the required reasoning the FRAMES questions give. Extractive models can't perform inference, they only extract a span that literally appears in the context they are given.

B3 and B4 shows that few-shot prompting improves accuracy with +4.9% over zero-shot prompting, even in this case with no retrieval. We hypothesize that few-shot conditions the model to produce shorter, and more direct answers rather than over-explaining and elaborating too much. However both models are limited by the LLM's parametric knowledge, and FRAMES is specifically designed to require retrieved evidence. These results are still useful to see a baseline and find out what the exact same LLM can do with and without retrieval.

Comparing B3(zero-shot LLM) with C1(single-step RAG) results show that adding a single-step BM25 retrieval nearly doubles the accuracy (+ 12.7%). This directly answers our core research question RQ1: RAG substantially improves QA over a capable LLM alone on this benchmark.

From C1 to C2 we can see that multi-hop retrieval adds +2.9% accuracy. The gain is modest but consistent with the retrieval recall improvement we saw (0.437 to 0.616). The modest answer accuracy gain relative to the much larger retrieval recall gain suggests that at this point retrieval is no longer the primary bottleneck, reasoning quality might now be a bigger constraint.

Token F1 and EM are misleading. B3 has Token F1 of 0.145, but the LLM-judge only gave it an accuracy of 0.138. Meanwhile C1 has a Token F1 of only 0.107, despite the much higher judge accuracy of 0.265. This is because the RAG model produces much longer explanatory answers. We can also see that EM is close to 0 for all LLM models.

6.2 Accuracy by Reasoning Type

We also have LLM-judge accuracy broken down by reasoning type.

Reasoning Type	B1	B2	B3	B4	C1	C2
Multiple constraints	0.004	0.081	0.136	0.199	0.265	**0.309**
Temporal reasoning	0.000	0.029	0.130	0.188	0.246	**0.261**
Numerical reasoning	0.007	0.027	0.108	0.108	0.176	**0.223**
Tabular reasoning	0.009	0.068	0.120	0.145	0.188	**0.222**
Post-processing	0.000	0.036	0.073	0.036	0.164	**0.200**

Table 5: LLM-Judge accuracy broken down by reasoning type.

Key observations:

Multiple constraints are the easiest category to answer. They often require finding two or more facts and connecting them, something which multi-hop retrieval handles well.

Post-processing is the hardest across all models. These questions often require transforming a retrieved fact (e.g., "how many years earlier would X have to be born to..."). Even with all the correct facts being retrieved, the model must still perform a computation. The FRAMES paper reports that around 80% of oracle failures are numerical/tabular/post-processing.

Few-shot helps for multiple constraints and temporal, but not post-processing. This indicates this may be because the few-shot examples did not include computation-heavy examples, and possibly also that the format conditioning could have pushed the model toward answering short and concise which skips the necessary calculation.

Generally we can also see that the advantage of RAG is consistent across all reasoning types. For every category we can see that C2 > C1 > B4 > B3, which confirms that retrieval helps regardless of the type of reasoning required.

6.3 Accuracy by Number of Required Articles

This result is reported specifically to test the "Lost in the Middle" hypothesis (Liu et al., 2023).

C1 shows a sharp accuracy drop beyond 3 required articles, which is expected given that it only retrieves 5 articles per question and can't cover the evidence needed for more complex questions. C2 is more robust in the 3–5 article range. At n=3, C2 reaches 0.324 compared to C1's 0.246, which demonstrates the value of iterative retrieval. Beyond 6 required articles, sample sizes become too small to draw reliable conclusions and the results are quite noisy. These patterns are consistent with the positional bias findings of (Liu et al., 2023): as more articles are retrieved, the relevant content risks being "lost in the middle" of the context window. C2 has an advantage at n=3-5 which could partly come from better retrieval coverage, but the positional bias effect could suppress it.

7 Error Analysis

Each question for each model is classified on two binary dimensions: Gold retrieved: at least one gold article title appears in the model's retrieved set Answer correct: based on if LLM-as-Judge gave correct or incorrect.

This gives us four cells:

	Gold retrieved	**Gold not retrieved**
Answer correct	[A] Both succeeded	[B] Parametric hit
Answer incorrect	[C] Reasoning failure	[D] Retrieval failure

Table 6: The 2×2 error matrix to distinguish retrieval failures from reasoning failures.

We will only be doing this error analysis on B2, C1 and C2, since those are the only models using retrieval. Results:

Key observations:

Model	A	B	C	D
B2	4.9%	1.9%	71.8%	21.4%
C1	21.6%	4.9%	55.1%	18.4%
C2	26.9%	2.4%	60.7%	10.0%

Table 7: Error analysis for the three retrieval-based models.

Retrieval failure (D) drops sharply from C1 -> C2 (18.4% -> 10.0%). Multi-hop retrieval successfully reduces the cases where the gold evidence is simply not found at all. This confirms the retrieval recall improvement translates to fewer pure retrieval failures.

Reasoning failure (C) remains the dominant failure mode in C1 and C2 (55.1% and 60.7%). Even when the gold articles are retrieved, the model can't produce the correct answer.

The reasoning failure rate is much higher than in the FRAMES paper (around 27%). The paper tested oracle retrieval where gold passages were directly provided to the model. Our reasoning failure rate at 69-72% reflects: (a) only partial gold retrieval (we retrieve at least 1 gold article, not necessarily all); (b) a smaller and less capable model; (c) the full article content instead of pre-extracted relevant passages.

Parametric hits (B) in B3 and B4 are at 13.8% and 18.7% with zero retrieval. These represent the model's parametric knowledge. As for C1 and C2 the B cell is much smaller (4.9% and 2.4%), because they are using retrieval.

B2's reasoning failure rate of 93.7% shows that extractive models can't reason across multi-document FRAMES questions.

This provides a clear diagnostic signal, because it isolates whether improvements stem from better retrieval or improved reasoning. Notably, the large proportion of errors in cell [C] suggests diminishing returns from further retrieval improvements alone.

8 Limitations

All model experiments were done on a stratified 50% subset of 412 questions, instead of the full 824 questions from the FRAMES set. This is due to computational constraints. All models were being tested on identical data, but results may differ depending on the seed used.

We believe that a stronger LLM would significantly increase the overall accuracy across mod-

els. Our current model, Qwen2.5-14b-Instruct, is a much weaker model than other state-of-the-art models that are used in related work. We have considered trying stronger models, but they would require too much computational power for our current hardware.

We also use the same model as both the generator and judge. This could potentially lead to some leniency bias where the judge may be more inclined to rate its own outputs as correct. We believe a stronger judge such as GPT-4o would provide a more rigorous evaluation, as used in the original FRAMES paper with Gemini Pro 1.5 (Krishna et al., 2025)

The retrieval pipeline used in this project relied only on BM25, which can't capture semantic similarity between a paraphrased question and an article title. Dense retrievers such as ColBERTv2 or a hybrid approach would likely improve retrieval coverage. For our models we retrieve full Wikipedia article text and use a sliding window, which introduces noise if we compare it to an approach using clean, pre-chunked passages which typically is the case in other RAG experiments.

Our multi-hop model is capped at three hops, while the FRAMES paper's multi-step baseline uses up to five. A possible C3 model was considered with 5 hops, but due to constraints on computational power and time this was not tested.

9 Ethical Considerations

One of the main benefits of RAG systems is that they increase trustworthiness to users by building their answer on trusted sources. However, RAG based systems are still susceptible to errors like any other Natural Language Processing model. Users still need to be critical of the generated answers instead of just accepting them as facts. Even when relevant articles are retrieved, the LLM still has the ability to ignore them and generate hallucinated answers. Although our error analysis directly addresses this issue, real users have no way of detecting whether or not a model generates answers based on the retrieved documents. This concern can be mitigated by letting users know that the answers the system gives could be wrong, similar to how ChatGPT does.

The document corpus' source being Wikipedia is also a credibility concern as many of the articles may contain biases from the writers that could end up skewing the answers. According to a community insight report by WikiMedia, 87 percent of Wikipedia contributors identify as male, with more than half living in Western countries. (Wikimedia Foundation, 2020). This means that topics related to this demographic tend to be covered in greater depth and more accurate than other articles. Because our RAG system retrieves these articles as evidence, it may inherit these biases directly.

10 Conclusion and Future Work

10.1 Does RAG Help?

Based on the results, we can conclude that RAG helps QA models substantially. The LLM-as-judge accuracy progressions are: B3 (13.8%) -> C1 (26.5%) -> C2 (29.4%). Single-step retrieval nearly doubles the accuracy. This directly answers the research question: RAG is the most impactful intervention. The gain from a single-step RAG (+12.7%) is much greater than the gain we achieved from few-shot prompting (+4.9%) or multi-hop retrieval over single hop (+2.9%).

However, we must also take note of the fact that all models perform far below the FRAMES paper's multi-step baseline (0.660) and oracle (0.729). This gap is likely due to differences in model capacity, retrieval quality, and the use of full-document context rather than curated passages.

10.2 Generative LLMs vs. Fine-Tuned Extractive Models

We can also conclude that generative LLMs are substantially better on FRAMES. Our B3 model (Qwen2.5-14B, no retrieval) has an accuracy of 13.8%, which is much higher than B2 (RoBERTa-SQuAD + retrieval), which only had 6.8+% accuracy. In other words, the LLM without retrieval outperforms the extractive model with retrieval.

This should not come as a surprise. The FRAMES questions require synthesizing information across multiple documents and to perform inference, which an extractive model cannot do. Extractive models are designed for reading comprehension where the answer is a literal span in a single passage. The RoBERTa model is fine-tuned on SQuAD, but not on FRAMES-style questions(Rajpurkar et al., 2016). Rather, one could try to test a fine-tuned generative model.

10.3 What Hurts Retrieval Performance?

Based on our results, we can conclude that there are three primary factors that can hurt retrieval

performance:

Factor 1: Multi-hop question structure. BM25 Recall@5 drops from 0.527 (2-article questions) to 0.198 (11-article questions). This means that a single query over the full question will not be able to fetch all the required articles. This makes more advanced solutions more desirable such as DSPy BasicMultiHop.

Factor 2: BM25 lexical mismatch. The retrieval method BM25 is simple to use, and not computationally heavy to run, but it has its weaknesses. BM25 relies on exact term overlap, meanwhile the questions in FRAMES are often paraphrases. Therefore one should definitely try use dense retrievers like ColBERTv2, which can also capture semantic similarity. One could also try DPR, but it might be risky as it has proven to have poor zero-shot generalization (-47.7% vs BM25 on BEIR). A simpler solution could be to combine using BM25 with a cross-encoder re-ranker.

Factor 3: Distractor contamination. Multi-hop retrieval can accumulate up to 15 articles, and this includes false positives. By false positives we mean articles that are related in terms of semantics and thus scores high on BM25, but do not contain the answer. (Cuconasu et al., 2024) show that this can reduce accuracy by up to -67%. This is also noticeable when comparing C2 to C1, which had a +2.9% accuracy increase, despite it having a huge recall improvement of 17.9%. This indicates that retrieving similar articles that are not helpful can really hurt performance, which can offset the benefit of better coverage.

10.4 The Reasoning Bottleneck

Even when retrieval succeeds, the reasoning failure rate of our C2 model is 69.3%. There can be several reasons behind this, and something that definitely would be worth exploring:

- The answer could require computations over retrieved facts, which our model potentially struggle with.
- The relevant fact could be present, but buried in 700-character passages mixed with text that is unrelated.
- The model might not be able to synthesize across all the accumulated articles at the same time.
- The judge could also sometimes be wrong.

This also aligns with the theory from (Li et al., 2025), where logic-based hallucinations (reasoning failures given correct facts) can't be fixed by better retrieval. Instead they might require reasoning-specific interventions such as chain-of-thought prompting, access to tools such as a calculator, or simply getting a stronger base model.

10.5 Metric Discussion

Token F1 and EM are poor primary metrics for this task, and there are two reasons:

- LLM responses can be long and explanatory. It can be reduced to a great degree by good prompting, but EM can still give 0 for correct answers.
- Short gold answers may have low token overlap with an answer that is correct, but in long-form.

Because of this the LLM-as-judge approach which was used in the FRAMES paper itself, and also validated by RAGAS, is what we see as the appropriate primary metric. For this project the main limitation is that we used the same model as a generator and judge, which potentially could introduce a bias on leniency, especially if the judge is not prompted strictly enough.

10.6 Future Work

We believe it would be extremely interesting to do an **oracle experiment**: providing the model with the gold Wikipedia articles directly for each question, bypassing retrieval entirely. This would establish the highest accuracy achievable for Qwen2.5-14B-Instruct. This would help us sharpen our evaluation and error analysis of the model, and provide a ceiling that is specific for our model.

One can also explore detection of hallucinated retrieval, which is cases where the model generates an answer that is not grounded in any of the articles retrieved. Here one could use RAGAS Faithfulness (Es et al., 2025), which measures the fraction of answer statements that can be verified against the retrieved context.

Beyond this, replacing the BM25 retriever with a dense or hybrid retriever is very likely to improve the retrieval of the models. Extending the retrieval loop to five hops or even more, in addition to using a stronger and independent judge model are also what we see as natural next steps to experiment with.

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A Reproducibility

All code, outputs, and the Wikipedia corpus are available at: https://github.com/Horvli/RAG_FRAMES

B Prompt Templates

All prompts were developed with the assistance of ChatGPT.

B.1 Zero-Shot Prompt (B3)

```
Answer the following question. Be concise
-- give only the answer, not a full
explanation.

Question: {question}
Answer:
```

B.2 Few-Shot Prompt (B4)

The following three examples are prepended to each question:

```
Answer questions concisely based on your
knowledge. Here are some examples:

Question: What is the capital city of the
country that hosts the FIFA World Cup most
recently before 2022?
Answer: Moscow

Question: How many Academy Awards has the
director of Inception won in total?
Answer: 0

Question: What year was the oldest
university in the United States founded?
Answer: 1636

Now answer:
Question: {question}
Answer:
```

B.3 Query Generation Prompt (C2)

Used at each hop to generate the next search query:

```
You are answering a complex question by
searching Wikipedia one step at a time.

Question: {question}

Information retrieved so far:
{context}

Write ONE short search query (5-10 words)
to find the next piece of information
needed.
```

```
Rules:
- On the first step, identify the first
  specific entity or fact to look up
- Each query must target ONE specific
  entity, person, event, or date
- Never paste the full question as a query
  -- break it into atomic lookups
- Do not repeat a previous query

Previous queries: {prev_queries}

Next search query (output the query only,
no explanation, no labels):
```

B.4 Answer Generation Prompt (C1, C2)

```
Answer the question using the Wikipedia
articles below as your primary source.
Read the articles carefully. If the answer
is in the articles, use it. If the articles
do not contain enough information, use your
best knowledge but keep the answer short.

{context}

Question: {question}

Answer (be concise -- one sentence or a
short phrase):
```

B.5 LLM-as-Judge Prompt

```
You are evaluating whether a model's answer
to a factual question is correct.

Question: {question}
Ground truth: {gold}
Model's answer: {predicted}

The model may include reasoning before
giving its answer. A response is CORRECT
if it contains the correct factual answer
anywhere in the text, even if surrounded
by explanation.
A response is INCORRECT if it:
- Never states the correct answer
- States the wrong fact as the answer
- Refuses to answer or says it cannot
  determine the answer
- Is vague or hedged without committing
  to a correct answer

Judge an answer based on the Ground truth
given only, it does not matter if you agree
```

```
with the explanation if the prediction
doesn't match the ground truth.
Respond with exactly one word:
CORRECT or INCORRECT.
```

C Additional Figures

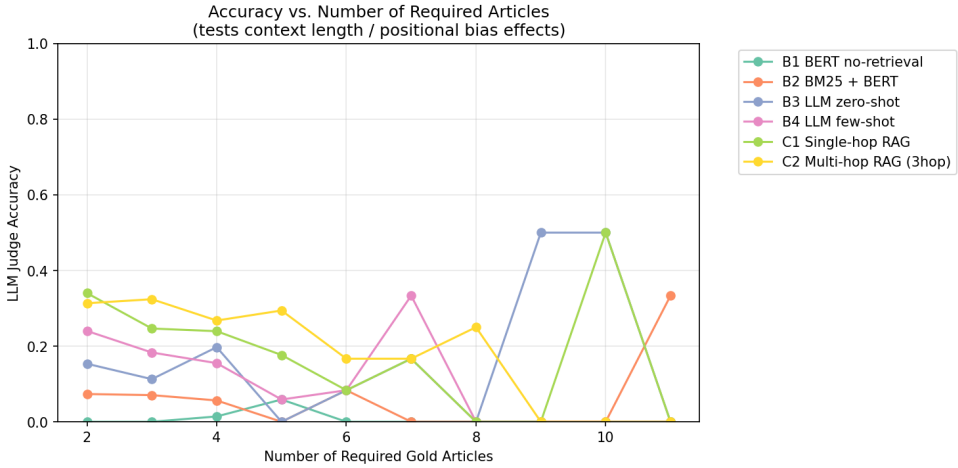

Figure 1: LLM-Judge accuracy by number of required gold articles for all six models

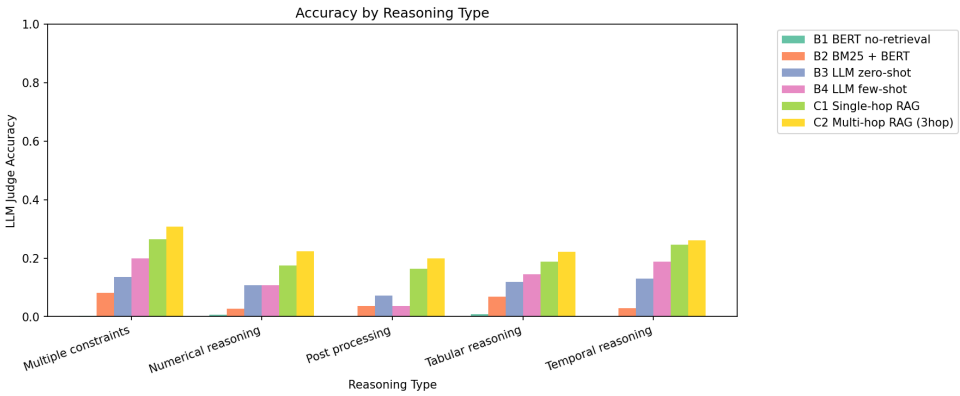

Figure 2: LLM-Judge accuracy by reasoning type for all six models.

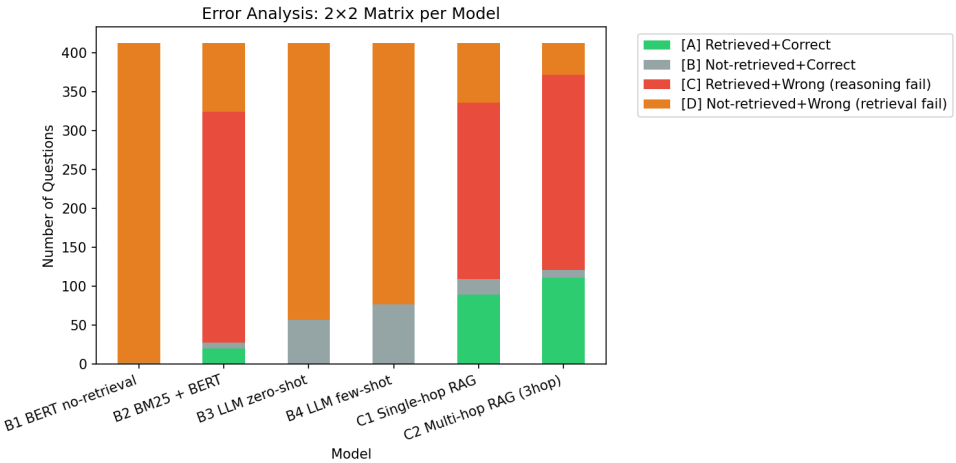

Figure 3: Error analysis 2×2 matrix breakdown per model, showing the distribution of retrieval and reasoning failures.